

Introduction To Software Engineering

What is Software Engineering?

- The term **software engineering** is the product of two words, **software**, and **engineering**.
- The **software** is a collection of integrated programs.
- Software subsists of carefully-organized instructions and code written by developers on any of various particular computer languages.
- Computer programs and related documentation such as requirements, design models and user manuals.
- Engineering** is the application of **scientific** and **practical** knowledge to **invent, design, build, maintain, and improve frameworks, processes, etc.**

- Software is a program or set of programs containing instructions which provide desired functionality . And Engineering is the processes of designing and building something that serves a particular purpose and find a cost effective solution to problems.
- **Engineering** on the other hand, is all about developing products, using well-defined, scientific principles and methods.

What is Software Engineering?

- **Software Engineering** is an engineering branch related to the evolution of software product using systematic well-defined scientific principles, techniques, and procedures.
- The result of software engineering is an effective and reliable software product.

What is Software Engineering?



Why is Software Engineering required?

Software Engineering is required due to the following reasons:

- To manage Large software
- For more Scalability
- Cost Management
- To manage the dynamic nature of software
- For better quality Management

Why is Software Engineering required?

The necessity of software engineering appears because of a higher rate of progress in user requirements and the environment on which the program is working.

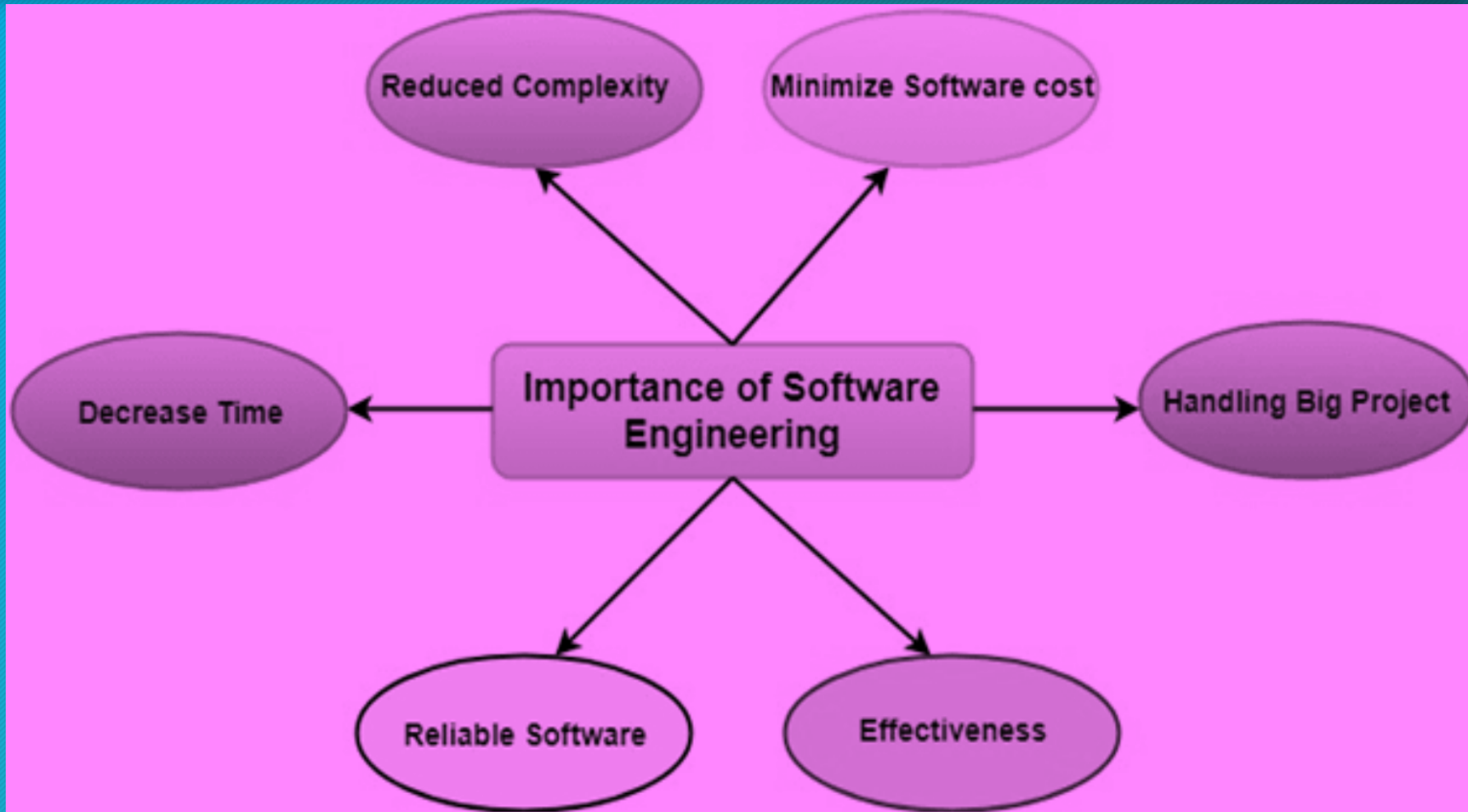
- **Huge Programming:** It is simpler to manufacture a wall than to a house or building, similarly, as the measure of programming become extensive engineering has to step to give it a scientific process.
- **Adaptability:** If the software procedure were not based on scientific and engineering ideas, it would be simpler to re-create new software than to scale an existing one.
- **Cost:** As the hardware industry has demonstrated its skills and huge manufacturing has let down the cost of computer and electronic hardware. But the cost of programming remains high if the proper process is not adapted.
- **Dynamic Nature:** The continually growing and adapting nature of programming hugely depends upon the environment in which the client works. If the quality of the software is continually changing, new upgrades need to be done in the existing one.
- **Quality Management:** Better procedure of software development provides a better and quality software product

Characteristics of a good software engineer

The features that good software engineers should possess are as follows:

- Exposure to systematic methods, i.e., familiarity with software engineering principles.
- Good technical knowledge of the project range (Domain knowledge).
- Good programming abilities.
- Good communication skills. These skills comprise of oral, written, and interpersonal skills.
- High motivation.
- Sound knowledge of fundamentals of computer science.
- Ability to work in a team
- Discipline

Importance of Software Engineering



Importance of Software Engineering

(1) Reduces complexity:

- Large software systems are always complicated and challenging to progress.
- Software engineering divides big problems into various small issues. And then start solving each small issue one by one.
- All these small problems are solved independently to each other and then integrated together to produce the software product.

(2) To minimize software cost:

- A lot of resources are required to develop large-scale software systems, such as: manpower, software licenses, hardware...etc.
- As companies seek to build cutting-edge software to drive growth, determining the overall budget becomes very tricky.
- Software engineering provides systematic means for having regular interaction and obtaining a budget estimates.

Importance of Software Engineering

(3) To decrease time:

- One of the main criteria to measuring project success and yet the most challenging is delivering software projects on time.
- Schedule issues are the main reason for conflicts on projects, especially during the second half of projects where the actual implementation takes place to produce the final working product.
- Software engineering involves the processes required to ensure timely completion of a project.

(4) Handling Big projects:

- Big projects are not done in a couple of days, and they need lots of work, planning, and management. And to invest six and seven months of any company, it requires heaps of planning, direction, testing, and maintenance.
- Companies provide many resources to the plan to be completed.
- So to handle a big project without any problem, the company has to go for software engineering methods.

Importance of Software Engineering

- (5) Reliable software:
 - Software reliability is the “probability that the software will execute for a particular period of time without failure”.
 - Software engineering provides models for software quality measurements and evaluations. Examples include models for estimating defects inserted and removed throughout the software lifecycle.
- (6) Effectiveness:
 - Effectiveness means getting the desired results/ doing the right thing!
 - From example: achieving missions and goals; generating satisfied customers; producing work of high quality.
 - This includes proper use of: communication, technology, organizational and individual knowledge, and resources.

Software Engineering vs. Computer Science

- Computer science is concerned with theory and fundamentals; This field involves the understanding and application of both abstract and concrete knowledge.
- Software engineering is a field largely concerned with the application of engineering processes to the creation, maintenance, and design of software for a variety of different purposes.

Software Engineering vs. System Engineering?

- System engineering is concerned with all aspects of computer-based systems development including hardware, software and process engineering.
- System engineers are involved in system specification, architectural design, integration and deployment.
- Software engineering is part of this process concerned with developing the software infrastructure, control, applications and databases in the system.
- Systems engineering is older than Software Engineering:
- Complex industrial systems such as trains and chemical plants.
- As the percentage of software systems has increased, software engineering techniques are finding their way into systems engineering.

Software engineering

- The economies of ALL developed nations are dependent on software.
- More and more systems are software controlled
- Software engineering is concerned with theories, methods and tools for professional software development.
- Expenditure on software represents a significant fraction of GNP in all developed countries

Frequently asked questions about software engineering

Question	Answer
What is software?	Computer programs and associated documentation. Software products may be developed for a particular customer or may be developed for a general market.
What are the attributes of good software?	Good software should deliver the required functionality and performance to the user and should be maintainable, dependable and usable.
What is software engineering?	Software engineering is an engineering discipline that is concerned with all aspects of software production.
What are the fundamental software engineering activities?	Software specification, software development, software validation and software evolution.
What is the difference between software engineering and computer science?	Computer science focuses on theory and fundamentals; software engineering is concerned with the practicalities of developing and delivering useful software.
What is the difference between software engineering and system engineering?	System engineering is concerned with all aspects of computer-based systems development including hardware, software and process engineering. Software engineering is part of this more general process.

Frequently asked questions about software engineering

Question	Answer
What are the key challenges facing software engineering?	Coping with increasing diversity, demands for reduced delivery times and developing trustworthy software.
What are the costs of software engineering?	Roughly 60% of software costs are development costs, 40% are testing costs. For custom software, evolution costs often exceed development costs.
What are the best software engineering techniques and methods?	While all software projects have to be professionally managed and developed, different techniques are appropriate for different types of system. For example, games should always be developed using a series of prototypes whereas safety critical control systems require a complete and analyzable specification to be developed. You can't, therefore, say that one method is better than another.
What differences has the web made to software engineering?	The web has led to the availability of software services and the possibility of developing highly distributed service-based systems. Web-based systems development has led to important advances in programming languages and software reuse.